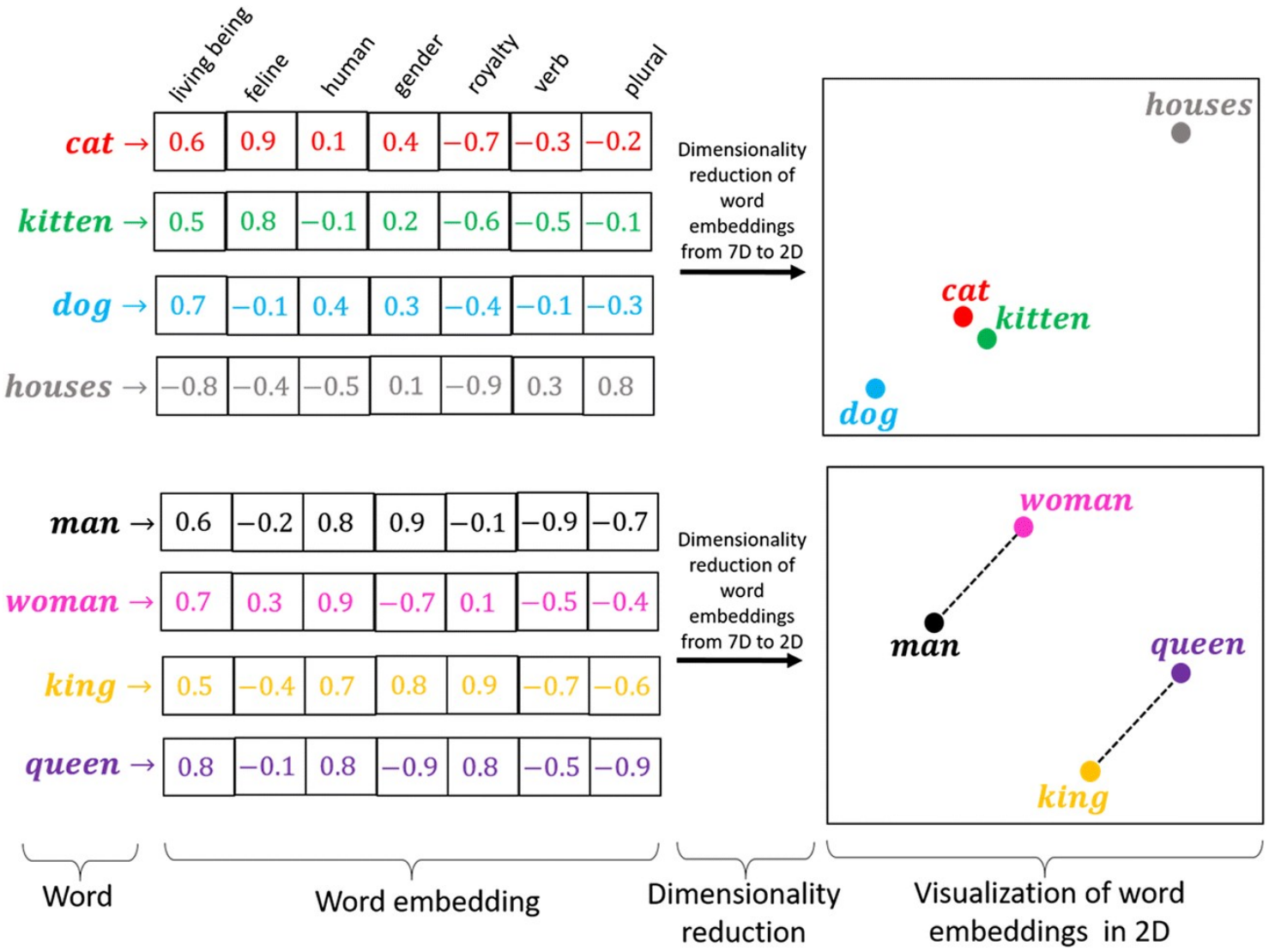
Retrieval Augmented Generation (RAG)

Embeddings

Dedicated AI model



ai-demos.dev

AI Demos

Vertex AI Embeddings API + Matching Engine demo

This demo features fast and scalable semantic search implemented with Vertex AI Embeddings API for Text, Multimodal Embeddings and Matching Engine. Embeddings API uses Large Language Models (LLMs) to generate embeddings representing text and image semantics, and Matching Engine provides a fully managed, high-scale, low latency vector database (a.k.a, approximate nearest neighbor search).

TRY THE DEMO | HOW WE BUILT IT

Get the closest matches based on your selected item.

STACKOVERFLOW (TEXT) | MERCARI TEXT-TO-IMAGE | MERCARI IMAGE-TO-IMAGE

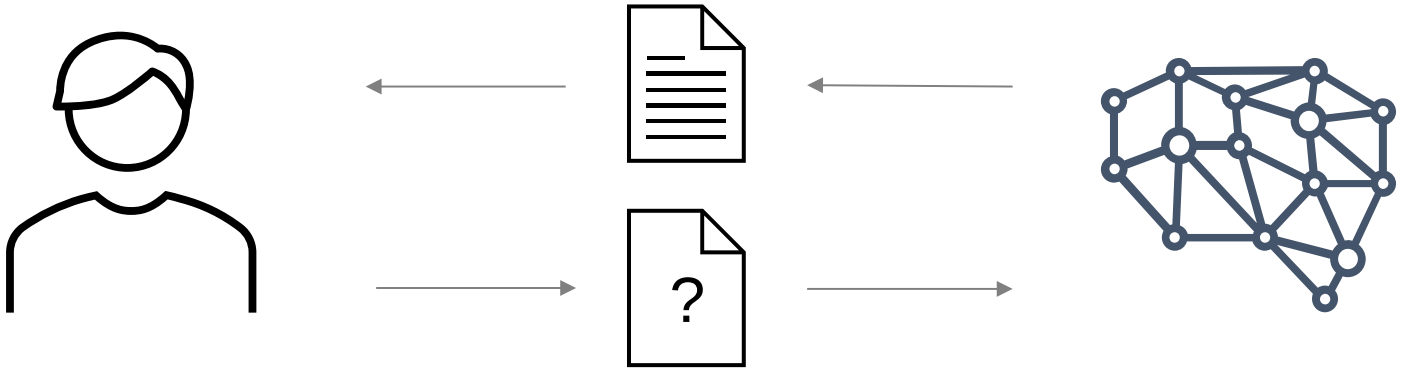
Questions from StackOverflow encoded using Vertex Text Embeddings.

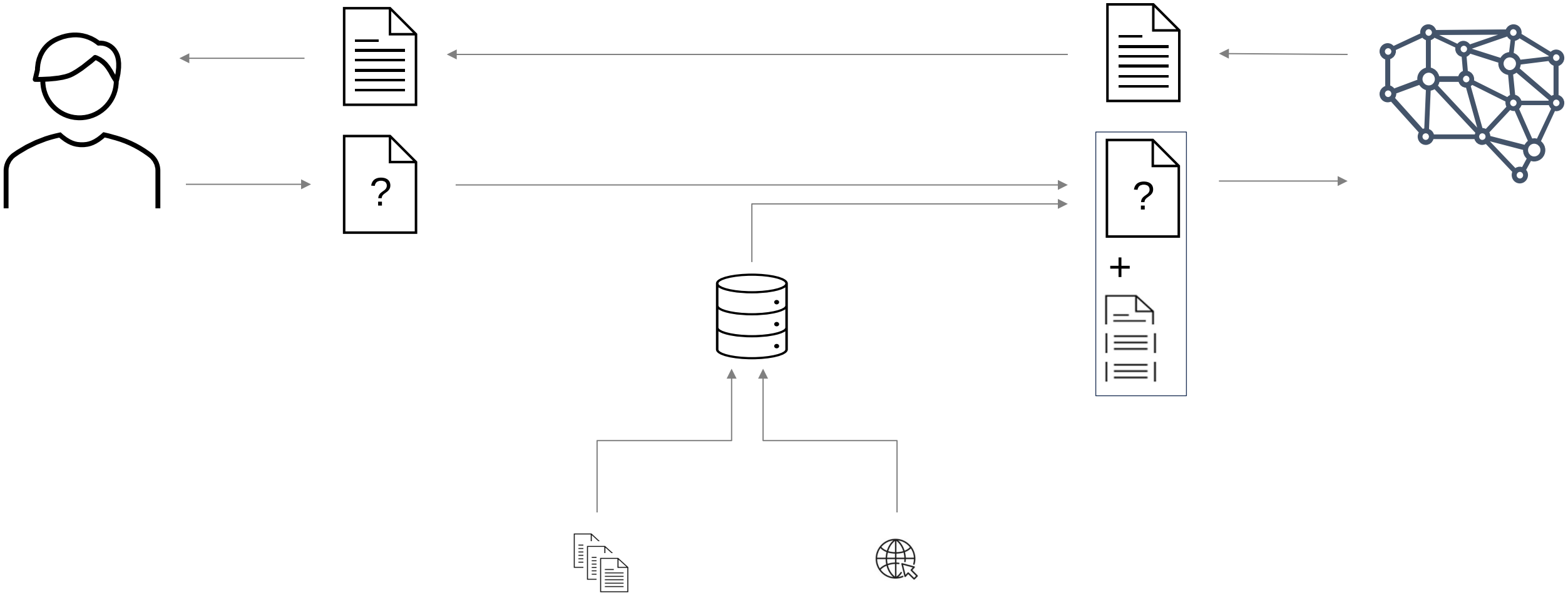
See code at Using Vertex AI Matching Engine and Vertex AI Embeddings for Text

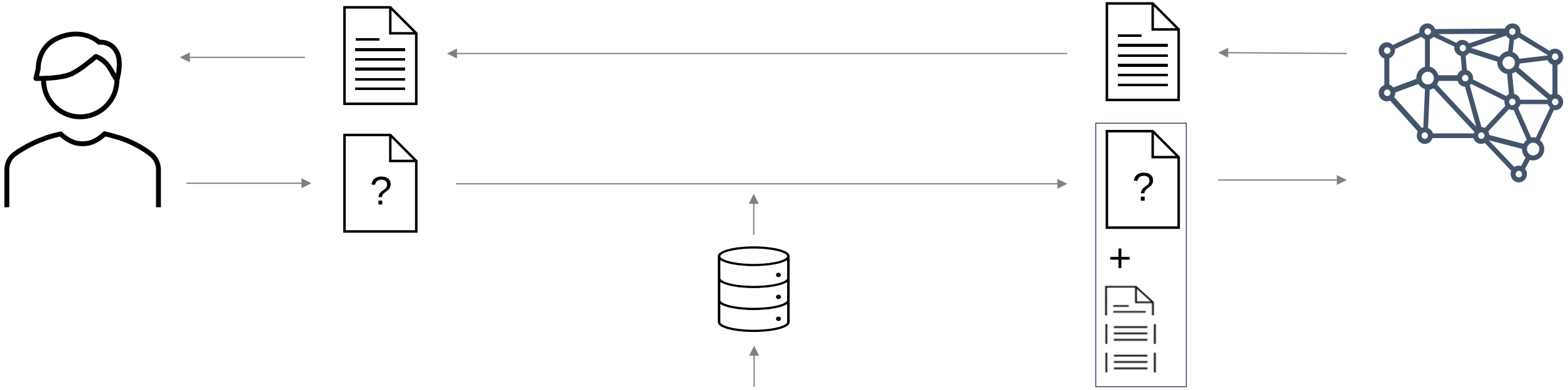
Search by text

Type a search query

Retrieval Augmented Generation (RAG)







indexing

- **Document 1**: "The quick brown fox jumps over the lazy dog."
- **Document 2**: "Lorem ipsum dolor sit amet, consectetur adipiscing elit."
- **Document 3**: "A quick brown dog outpaces a fast fox."

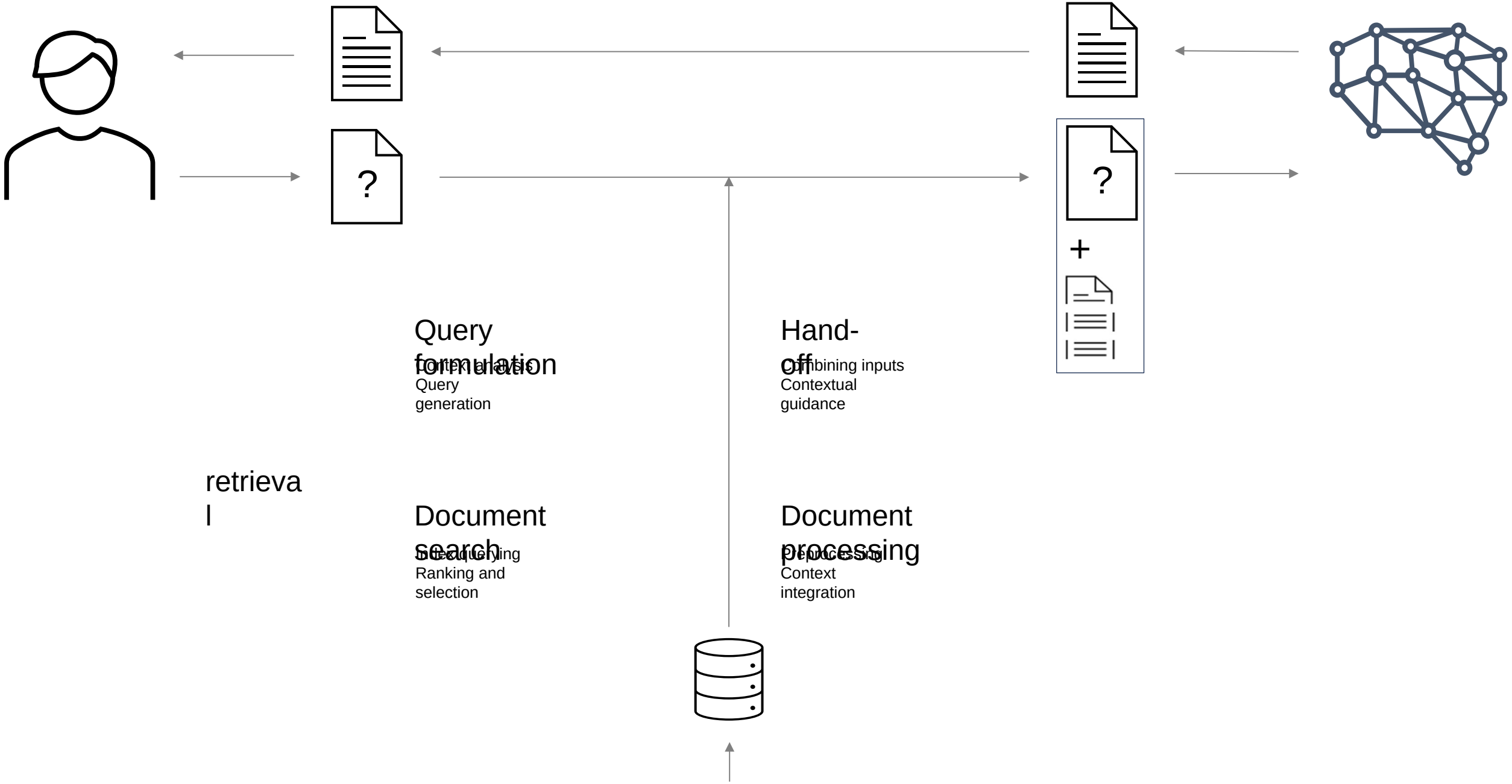
- **the**: [1]
- **quick**: [1, 3]
- **brown**: [1, 3]
- **fox**: [1, 3]
- **jumps**: [1]
- **over**: [1]
- **lazy**: [1]

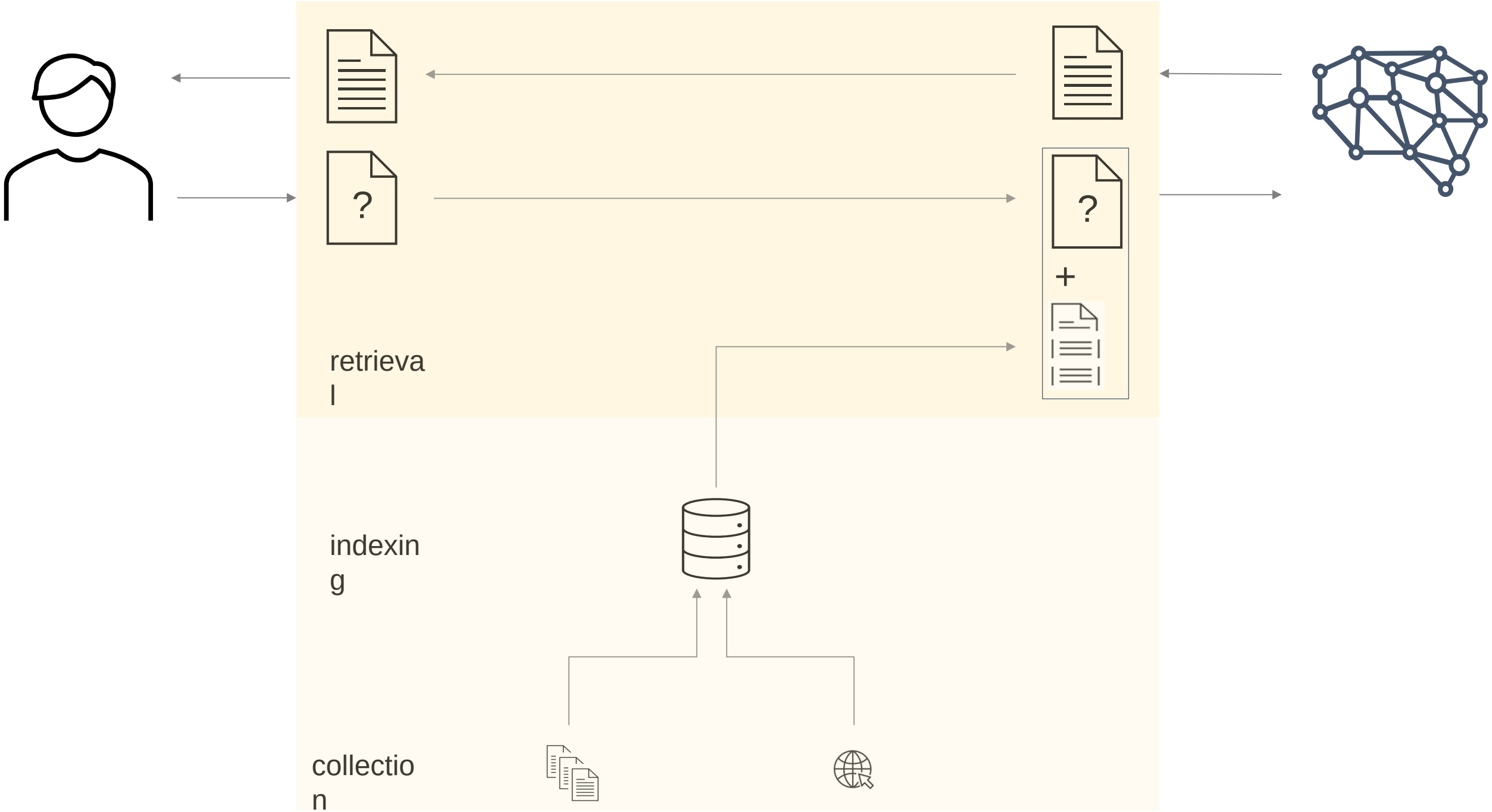
embedding

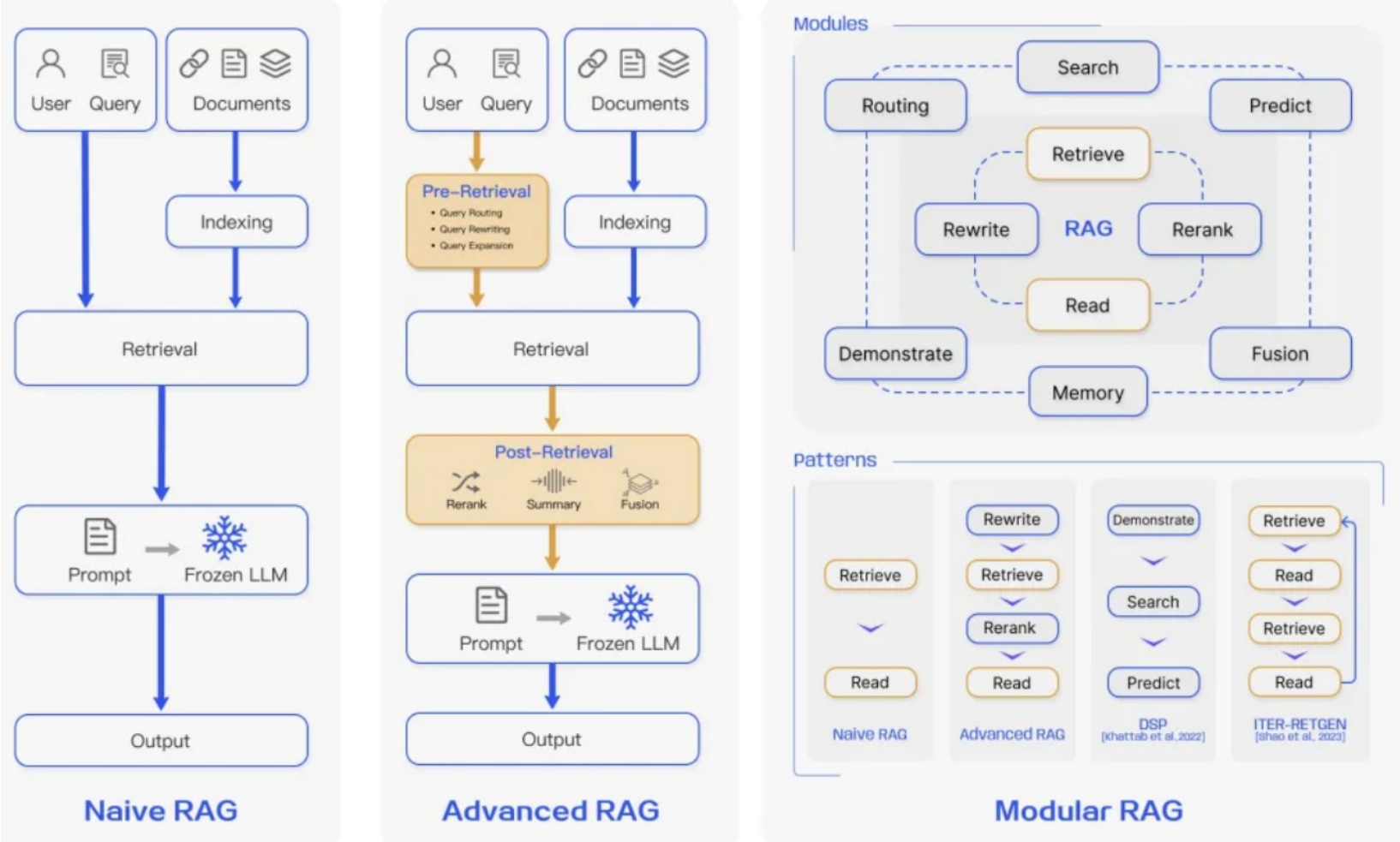
0.1	-0.7	1.5	-0.2	0.4	...
0.4	-0.1	0.5	-0.7	0.1	...
-0.3	0.2	1.3	0.3	0.5	...
0.1	-0.7	1.5	-0.2	0.4	...
0.4	-0.1	0.5	-0.7	0.1	...
-0.3	0.2	1.3	0.3	0.5	...

chunking

collection


?
?
+
Query formulation
Context analysis
Query generation
Hand-off
Combining inputs
Contextual guidance
retrieval
Document search
Indexquerying
Ranking and selection
Document processing
Preprocessing
Context integration


?
?
+
retrieva
l
indexin
g
collectio
n


User
Query
Documents
Indexing
Retrieval
Prompt
Frozen LLM
Output
Naive RAG
User
Query
Documents
Pre-Retrieval
• Query Routing
• Query Rewriting
• Query Expansion
Indexing
Retrieval
Post-Retrieval
Rerank
Summary
Fusion
Prompt
Frozen LLM
Output
Advanced RAG
Modules
Search
Routing
Predict
Retrieve
Rewrite
RAG
Rerank
Read
Demonstrate
Fusion
Memory
Patterns
Retrieve
Read
Naive RAG
Rewrite
Retrieve
Rerank
Read
Advanced RAG
Demonstrate
Search
Predict
DSP
[Khattab et al.,2022]
Retrieve
Read
Retrieve
Read
ITER-RETGEN
[Shao et al., 2023]
Modular RAG

RAG
Retrieval Augmented Generation

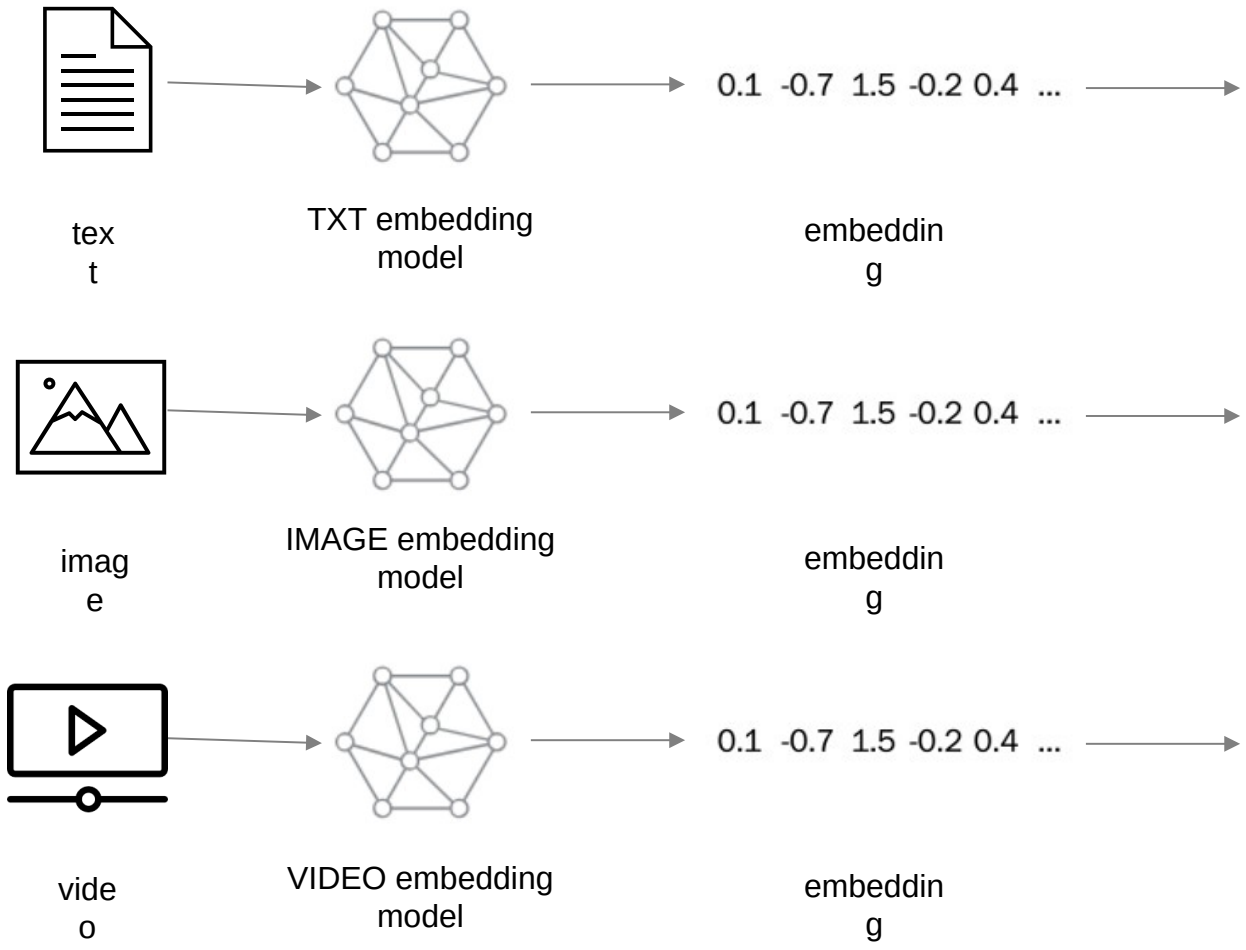
LangChain

LlamaIndex

Key libraries

Vector databases

In the retrieval process we compare vectors of embeddings



0.1 -0.7 1.5 -0.2 0.4 ...
0.4 -0.1 0.5 -0.7 0.1 ...
-0.3 0.2 1.3 0.3 0.5 ...
0.1 -0.7 1.5 -0.2 0.4 ...
0.4 -0.1 0.5 -0.7 0.1 ...
-0.3 0.2 1.3 0.3 0.5 ...
0.1 -0.7 1.5 -0.2 0.4 ...
0.4 -0.1 0.5 -0.7 0.1 ...
-0.3 0.2 1.3 0.3 0.5 ...
0.1 -0.7 1.5 -0.2 0.4 ...
0.4 -0.1 0.5 -0.7 0.1 ...
-0.3 0.2 1.3 0.3 0.5 ...
0.1 -0.7 1.5 -0.2 0.4 ...
0.4 -0.1 0.5 -0.7 0.1 ...
-0.3 0.2 1.3 0.3 0.5 ...

Search based on vector similarity

Vector databases

In the retrieval process we compare vectors of embeddings

Milvus
zilliz
Pinecone
vespa
marqo
drant
Weaviate
Chroma
Vald
LanceDB

Vector databases

> In the retrieval process we compare vectors of embeddings

docs.llamaindex.ai/en/stable/module_guides/storing/vector_stores.html

LlamaIndex 🦙
v0.10.19

Search ⌘ K

GETTING STARTED

Installation and Setup

How to read these docs

Starter Tutorial

High-Level Concepts

Customization Tutorial

Discover LlamaIndex Video Series

USE CASES

Q&A

Chatbots

Agents

Structured Data Extraction

Multi-modal

UNDERSTANDING

Building an LLM application

Using LLMs

Loading Data (Ingestion)

Indexing

Storing

Querying

Putting It All Together

Tracing and Debugging

Vector Stores

Vector stores contain embedding vectors of ingested document chunks (and sometimes the document chunks as well).

Simple Vector Store

By default, LlamaIndex uses a simple in-memory vector store that's great for quick experimentation. They can be persisted to (and loaded from) disk by calling `vector_store.persist()` (and `SimpleVectorStore.from_persist_path(...)` respectively).

Vector Store Options & Feature Support

LlamaIndex supports over 20 different vector store options. We are actively adding more integrations and improving feature coverage for each.

Vector Store	Type	Metadata Filtering	Hybrid Search	Delete	Store Documents	Async
Apache Cassandra®	self-hosted / cloud	✓		✓	✓	
Astra DB	cloud	✓		✓	✓	
Azure Cognitive Search	cloud		✓	✓	✓	
Azure CosmosDB MongoDB	cloud			✓	✓	
BaiduVectorDB	cloud	✓	✓		✓	
ChatGPT Retrieval Plugin	aggregator			✓	✓	
Chroma	self-hosted	✓		✓	✓	